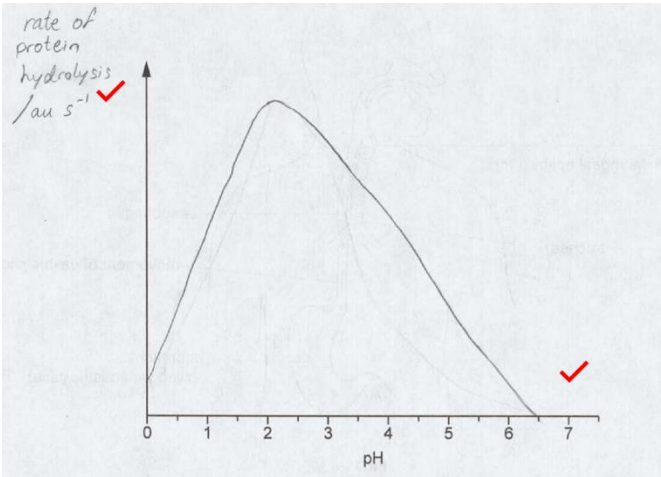


1(a)(i)	<i>any one from:</i> 1 calibrate colorimeter ; 2 use, a blank / AW, to set (colorimeter absorbance) to zero ;	1
1(a)(ii)	1 <i>rate of change in absorbance at pH 2.0</i> = $0.0036 / 3.6 \times 10^{-3}$; 2 <i>working:</i> $(0.52 - 0.23) / (120 - 40)$ or $0.29 / 80$; 3 <i>units:</i> au s^{-1} ;	3

Question	Answer	Marks												
1(b)(i)	any seven from: 1 prepare (at least) five solutions with stated pH values between pH 0 and 6.5 ; 2 same / stated, volume / concentration, of pepsin (solution) used ; 3 <u>method</u> to maintain a, constant / stated, temperature ; 4 equilibrate / AW, <u>all</u> (pepsin, pH / buffer / albumen) solutions separately (before mixing) ; 5 <u>method</u> to mix, pepsin and pH (buffer) albumen (solutions, before adding to colorimeter) ; 6 at each pH, measure / note / record / AW, absorbance at, set / stated, time intervals ; 7 use at least three replicates for each pH <u>and</u> calculate a mean <u>rate</u> (of change in absorbance / of protein hydrolysis, for each pH) ; 8 <u>method</u> to identify the optimum pH ; 9 <i>idea that</i> repeat experiment with smaller pH intervals around the optimum pH ; 10 safety comment with named hazard <u>and</u> risk <u>and</u> precaution ; <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> </thead> <tbody> <tr> <td>albumen / eggs</td><td>allergy / irritant</td><td>gloves / goggles / mask / PPE</td></tr> <tr> <td>pH buffer</td><td>irritant / corrosive (at pH 0, 1, 2)</td><td>gloves / goggles / mask / PPE</td></tr> <tr> <td>pepsin</td><td>allergy / irritant</td><td>gloves / goggles / mask / PPE</td></tr> </tbody> </table>	hazard	risk	precaution	albumen / eggs	allergy / irritant	gloves / goggles / mask / PPE	pH buffer	irritant / corrosive (at pH 0, 1, 2)	gloves / goggles / mask / PPE	pepsin	allergy / irritant	gloves / goggles / mask / PPE	7
hazard	risk	precaution												
albumen / eggs	allergy / irritant	gloves / goggles / mask / PPE												
pH buffer	irritant / corrosive (at pH 0, 1, 2)	gloves / goggles / mask / PPE												
pepsin	allergy / irritant	gloves / goggles / mask / PPE												

Question	Answer	Marks
1(b)(ii)	1 correct y-axis label ; e.g. rate of protein hydrolysis / au s^{-1} 2 appropriately shaped line ; 	2

Question	Answer	Marks
1(c)(i)	any two from: 1 test condition 2, decreases / hydrolyses / AW, CA3 <u>and</u> Sep70 ; 2 <i>idea that</i> acid / pH 4.0, with no active pepsin, does not change CA3 <u>and</u> Sep70 ; 3 (mass / size / AW of) CA3 is <u>smaller</u> than Sep70 ; ora	2
1(c)(ii)	any two from: 1 <i>idea that</i> pepsin only causes damage (to the laryngeal epithelium) if (hydrochloric) acid / pH 4.0, is also present ; 2 investigation, only carried out once / was not repeated ; 3 <i>idea that</i> no information on species of mammal used (in investigation) or <i>idea that</i> investigation, not carried out in a person / carried out in a laboratory ; 4 <i>idea that</i> a reduction in quantity of Sep70 and CA3 may not cause damage to the laryngeal epithelium ;	2